We will speak only minimally of philosophy (and not at all about suicide and the meaning of life). But in our scientific quest to solve the mysteries of space and time, we will be unrestrained. From the universe's smallest speck and earliest moments to its farthest reaches and most distant future, we will examine space and time in environments familiar and far-flung, with an unflinching eye seeking their true nature. As the story of space and time has yet to be fully written, we won't arrive at any final assessments. But we will encounter a series of developments—some intensely strange, some deeply satisfying, some experimentally verified, some thoroughly speculative—that will show how close we've come to wrapping our minds around the fabric of the cosmos and touching the true texture of reality.

Classical Reality

Historians differ on exactly when the modern scientific age began, but certainly by the time Galileo Galilei, René Descartes, and Isaac Newton had had their say, it was briskly under way. In those days, the new scientific mind-set was being steadily forged, as patterns found in terrestrial and astronomical data made it increasingly clear that there is an order to all the comings and goings of the cosmos, an order accessible to careful reasoning and mathematical analysis. These early pioneers of modern scientific thought argued that, when looked at the right way, the happenings in the universe not only are explicable but predictable. The power of science to foretell aspects of the future—consistently and quantitatively—had been revealed.

Early scientific study focused on the kinds of things one might see or experience in everyday life. Galileo dropped weights from a leaning tower (or so legend has it) and watched balls rolling down inclined surfaces; Newton studied falling apples (or so legend has it) and the orbit of the moon. The goal of these investigations was to attune the nascent scientific ear to nature's harmonies. To be sure, physical reality was the stuff of experience, but the challenge was to hear the rhyme and reason behind the rhythm and regularity. Many sung and unsung heroes contributed to the rapid and impressive progress that was made, but Newton stole the show. With a handful of mathematical equations, he synthesized everything known about motion on earth and in the heavens,